

Homework 5: Full-Stack CRUD with Node, Express, & MongoDB

Course: CSCI 39548 - Practical Web Development **Due Date:** May 18

1. Introduction

Now that we have covered the basics of building an API in class (refer back to [Lesson 14](#)), it is time to put those skills into practice. You have seen how Node.js and Express can handle routing, and how MongoDB can be used for persistent data storage.

For this assignment, you will build a simple Full-Stack CRUD (Create, Read, Update, Delete) application. This will serve as the foundational backend architecture for your final projects. You can use this excellent tutorial as a reference guide: [Building a Simple CRUD App with Express and MongoDB](#).

2. The Assignment Task

You are tasked with building a web application that allows users to manage a simple list of items. You can choose the theme (e.g., a list of favorite quotes, a task manager, a book library, or a character roster).

Technical Requirements

Your application must implement all four CRUD operations using Node, Express, and MongoDB:

1. **Create (POST):**
 - Provide an HTML form for users to input new data (e.g., a name and a quote).
 - Submit this form to an Express route that saves the new document to your MongoDB database.
2. **Read (GET):**
 - Fetch the data from your MongoDB database and render it on the page. (You can use a templating engine like EJS, as shown in the tutorial, or send JSON to a vanilla JS frontend).
3. **Update (PUT):**
 - Implement functionality to modify an existing entry.
 - This typically involves a button that triggers a fetch request with the PUT method to update the document in MongoDB.
4. **Delete (DELETE):**
 - Provide a button for each entry that allows the user to delete it.
 - This should trigger a fetch request with the DELETE method, removing the document from the database and refreshing the UI.

3. Submission Instructions

By **May 18**, please submit the following to this portal:

1. **GitHub Repository Link:** Your codebase must be pushed to a public GitHub repository. Please ensure you include a .gitignore file so you do not push your node_modules folder!
2. **MongoDB Connection:** Please ensure your MongoDB connection string uses environment variables (.env) to keep your credentials secure. Do **not** hardcode your database password into your server file.

Good luck, and remember to check the class notes from Lesson 14 if you get stuck on the Express routing!